for some k, $0 < k < n$ we have $\langle ku \rangle \leqslant \frac{1}{n}$, and deduce that there exists a sequence $q_n \to \infty$ with $q_n \langle q_n u \rangle < 1$.

Exercise 3.1.4. Extend Proposition 3.3 in the following way. Given u as in (3.10), and the nth convergent $\frac{p_n}{q_n}$, the $(n+1)$th convergent $\frac{p_{n+1}}{q_{n+1}}$ is characterized by being the ratio of the unique pair of positive integers (p_{n+1}, q_{n+1}) for which $|p_{n+1} - q_{n+1}u| < |p_n - q_n u|$ with $q_{n+1} > q_n$ minimal. Notice that the same cannot be said when using the expression $\left|u - \frac{p_n}{q_n}\right|$, as becomes clear in the case where $u > \frac{1}{3}$ is very close to $\frac{1}{3}$, in which case the first approximation is not $\frac{1}{2}$.

Exercise 3.1.5. Let $u = [a_0; a_1, \dots]$ with convergents $\frac{p_n}{q_n}$. Show that

$$\frac{1}{2q_{n+1}} \leqslant |p_n - q_n u| < \frac{1}{q_{n+1}}.$$

3.2 The Continued Fraction Map and the Gauss Measure

Let $Y = [0,1] \smallsetminus \mathbb{Q}$, and define a map $T : Y \to Y$ by

$$T(x) = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor,$$

where $\lfloor t \rfloor$ denotes the greatest integer less than or equal to t. Thus $T(x)$ is the fractional part $\left\{\frac{1}{x}\right\}$ of $\frac{1}{x}$. The graph of this so-called *continued fraction* or *Gauss map* is shown in Fig. 3.1.

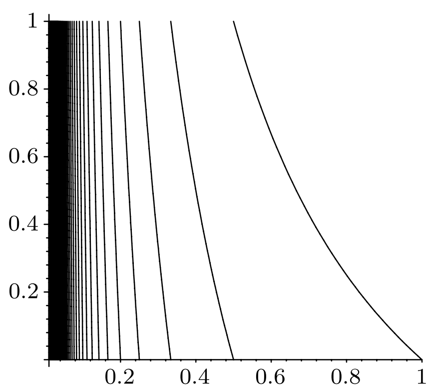

Fig. 3.1 The Gauss map

Gauss observed in 1845 that T preserves[(40)] the probability measure given by

$$\mu(A) = \frac{1}{\log 2}\int_A \frac{1}{1+x}\,\mathrm{d}x,$$

by showing that the Lebesgue measure of $T^{-n}I$ converges to $\mu(I)$ for each interval I.

This map will be studied via a geometric model (for its invertible extension) in Chap. 9; in this section we assemble some basic facts from an elementary point of view, showing that the Gauss measure is T-invariant and ergodic. Since the measure defined in Lemma 3.5 is non-atomic, we may extend the map to include the points 0 and 1 in any way without affecting the measurable structure of the system.

Lemma 3.5. *The continued fraction map* $T(x) = \left\{\frac{1}{x}\right\}$ *on* $(0,1)$ *preserves the Gauss measure* μ *given by*

$$\mu(A) = \frac{1}{\log 2}\int_A \frac{1}{1+x}\,\mathrm{d}x$$

for any Borel measurable set $A \subseteq [0,1]$.

A geometric and less formal proof of this will be given on page 93 using basic properties of the invertible extension of the continued fraction map in Proposition 3.15.

Proof of Lemma 3.5. It is sufficient to show that $\mu\left(T^{-1}[0,s]\right) = \mu\left([0,s]\right)$ for every $s > 0$. Clearly

$$T^{-1}[0,s] = \{x \mid 0 \leqslant T(x) \leqslant s\} = \bigsqcup_{n=1}^{\infty}\left[\frac{1}{s+n}, \frac{1}{n}\right]$$

is a disjoint union. It follows that

$$\begin{aligned}
\mu\left(T^{-1}[0,s]\right) &= \frac{1}{\log 2}\sum_{n=1}^{\infty}\int_{1/(s+n)}^{1/n}\frac{1}{1+x}\,\mathrm{d}x\\
&= \frac{1}{\log 2}\sum_{n=1}^{\infty}\Bigl(\log(1+\tfrac{1}{n}) - \log(1+\tfrac{1}{s+n})\Bigr)\\
&= \frac{1}{\log 2}\sum_{n=1}^{\infty}\Bigl(\log(1+\tfrac{s}{n}) - \log(1+\tfrac{s}{n+1})\Bigr) \qquad (3.16)\\
&= \frac{1}{\log 2}\sum_{n=1}^{\infty}\int_{s/(n+1)}^{s/n}\frac{1}{1+x}\,\mathrm{d}x\\
&= \mu\left([0,s]\right),
\end{aligned}$$

completing the proof. The identity used in (3.16) amounts to

$$\frac{1+\frac{s}{n}}{1+\frac{s}{n+1}} = \frac{1+\frac{1}{n}}{1+\frac{1}{s+n}},$$

which may be seen by multiplying numerator and denominator of the left-hand side by $\frac{n+1}{n+s}$, and the interchange of integral and sum is justified by absolute convergence. □

Thus Lemma 3.5 shows that $\left([0,1], \mathscr{B}_{[0,1]}, \mu, T\right)$ is a measure-preserving system.

Define for $x \in Y = [0,1] \smallsetminus \mathbb{Q}$ and $n \geqslant 1$ the sequence of natural numbers $(a_n) = (a_n(x))$ by

$$\frac{1}{1+a_n} < T^{n-1}(x) < \frac{1}{a_n}, \tag{3.17}$$

or equivalently by

$$a_n(x) = \left\lfloor \frac{1}{T^{n-1}x} \right\rfloor \in \mathbb{N}. \tag{3.18}$$

For any sequence $(a_n)_{n\geqslant 1}$ of natural numbers we define the continued fraction $[a_1, a_2, \dots]$ just as in (3.1) with $a_0 = 0$.

Lemma 3.6. *For any irrational $x \in [0,1] \smallsetminus \mathbb{Q}$ the sequence $(a_n(x))$ defined in (3.18) gives the digits of the continued fraction expansion to x. That is,*

$$x = [a_1(x), a_2(x), \dots].$$

Proof. Define $a_n = a_n(x)$ and let $u = [a_1, a_2, \dots]$ be the limit as in (3.10) with $a_0 = 0$. By (3.11) we have

$$\frac{p_{2n}}{q_{2n}} < u < \frac{p_{2n+1}}{q_{2n+1}}$$

and by (3.8) and the inequality (3.6) we have

$$\frac{p_{2n+1}}{q_{2n+1}} - \frac{p_{2n}}{q_{2n}} = \frac{1}{q_{2n}q_{2n-1}} \leqslant \frac{1}{2^{2n-2}}.$$

We now show by induction that

$$[a_1, \dots, a_{2n}] = \frac{p_{2n}}{q_{2n}} < x < \frac{p_{2n+1}}{q_{2n+1}} = [a_1, \dots, a_{2n+1}], \tag{3.19}$$

which together with the above shows that $u = x$.

Recall that $\frac{p_0}{q_0} = 0$ and $\frac{p_1}{q_1} = \frac{1}{a_1}$, so (3.19) holds for $n = 0$ because of the definition of a_1 in (3.18). Now assume that the inequality (3.19) holds for a given n and all $x \in [0,1]$. In particular, we may apply it to $T(x)$ to get

$$[a_2, \dots, a_{2n+1}] < T(x) < [a_2, \dots, a_{2n+2}].$$

Since $T(x) = \frac{1}{x} - a_1$ we get

$$a_1 + [a_2, \dots, a_{2n+1}] < \frac{1}{x} < a_1 + [a_2, \dots, a_{2n+2}]$$

and therefore

$$[a_1, \dots, a_{2n+2}] = \frac{1}{a_1 + [a_2, \dots, a_{2n+2}]} < x,$$

$$x < \frac{1}{a_1 + [a_2, \dots, a_{2n+1}]} = [a_1, \dots, a_{2n+1}]$$

as required. □

This gives a description of the continued fraction map as a shift map: the list of digits in the continued fraction expansion of $x \in [0,1] \smallsetminus \mathbb{Q}$ defines a unique element of $\mathbb{N}^{\mathbb{N}}$, and the diagram

$$\begin{tikzcd} \mathbb{N}^{\mathbb{N}} \arrow[r, "\sigma"] \arrow[d] & \mathbb{N}^{\mathbb{N}} \arrow[d] \\ (0,1) \arrow[r, "T"'] & (0,1) \end{tikzcd}$$

commutes, where σ is the left shift and the vertical map sends a sequence of digits $(a_n)_{n \geqslant 1}$ to the real irrational number defined by the continued fraction expansion.

In Corollary 3.8 we will draw some easy consequences[(41)] of ergodicity for the Gauss measure μ in terms of properties of the continued fraction expansion for almost every real number. Given a continued fraction expansion, recall that the *convergents* are the terms of the sequence of rationals $\frac{p_n(x)}{q_n(x)}$ in lowest terms defined by

$$\frac{p_n(x)}{q_n(x)} = \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \dots + \cfrac{1}{a_n}}}}.$$

Theorem 3.7. *The continued fraction map $T(x) = \{\frac{1}{x}\}$ on $(0,1)$ is ergodic with respect to the Gauss measure μ.*

Before proving this[(42)] we develop some more of the basic identities for continued fractions. Given a continued fraction expansion $u = [a_0; a_1, \dots]$ of an irrational number u, we write $u_n = [a_n; a_{n+1}, \dots]$ for the nth tail of the expansion. By Lemma 3.1 applied twice, we have

$$\begin{pmatrix} p_{n+k} \\ q_{n+k} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_{n+k} & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$= \begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} \begin{pmatrix} a_{n+1} & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_{n+k} & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Writing $p_k(u_{n+1})$ and $q_k(u_{n+1})$ for the numerator and denominator of the kth convergents to u_{n+1}, we can apply Lemma 3.1 again to deduce that

$$\begin{pmatrix} p_{n+k} \\ q_{n+k} \end{pmatrix} = \begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} \begin{pmatrix} p_{k-1}(u_{n+1}) & p_{k-2}(u_{n+1}) \\ q_{k-1}(u_{n+1}) & q_{k-2}(u_{n+1}) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

so

$$\frac{p_{n+k}}{q_{n+k}} = \frac{p_n \frac{p_{k-1}(u_{n+1})}{q_{k-1}(u_{n+1})} + p_{n-1}}{q_n \frac{p_{k-1}(u_{n+1})}{q_{k-1}(u_{n+1})} + q_{n-1}},$$

which gives

$$u = \frac{p_n u_{n+1} + p_{n-1}}{q_n u_{n+1} + q_{n-1}} \tag{3.20}$$

in the limit as $k \to \infty$. Notice that the above formulas are derived for a general positive irrational number u. If $u = [a_1, \dots] \in (0,1)$, then $u_{n+1} = (T^n(u))^{-1}$ so that

$$u = \frac{p_n + p_{n-1}T^n(u)}{q_n + q_{n-1}T^n(u)}. \tag{3.21}$$

Proof of Theorem 3.7. The description of the continued fraction map as a shift on the space $\mathbb{N}^{\mathbb{N}}$ described above suggests the method of proof: the measure μ corresponds to a rather complicated measure on the shift space, but if we can control the measure of cylinder sets (and their intersections) well enough then we may prove ergodicity along the lines of the proof of ergodicity for Bernoulli shifts in Proposition 2.15. For two expressions f, g we write $f \asymp g$ to mean that there are absolute constants $C_1, C_2 > 0$ such that

$$C_1 f \leqslant g \leqslant C_2 f.$$

Given a vector $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{N}^n$ of length $|\mathbf{a}| = n$, define a set

$$I(\mathbf{a}) = \{[x_1, x_2, \dots] \mid x_i = a_i \text{ for } 1 \leqslant i \leqslant n\}$$

(which may be thought of as an interval in $(0,1)$, or as a cylinder set in $\mathbb{N}^{\mathbb{N}}$).

The main step towards the proof of the theorem is to show that

$$\mu\left(T^{-n}A \cap I(\mathbf{a})\right) \asymp \mu(A)\mu(I(\mathbf{a})) \tag{3.22}$$

for any measurable set A. Notice that for the proof of (3.22) it is sufficient to show it for any interval $A = [d, e]$; the case of a general Borel set then follows by a standard approximation argument (the set of Borel sets satisfying (3.22) with a fixed choice of constants is easily seen to be a monotone class, so Theorem A.4 may be applied).

Now define $\frac{p_n}{q_n} = [a_1, \dots, a_n]$ and $\frac{p_{n-1}}{q_{n-1}} = [a_1, \dots, a_{n-1}]$. Then $u \in I(\mathbf{a})$ if and only if $u = [a_1, \dots, a_n, a_{n+1}(u), \dots]$, and so $u \in I(\mathbf{a}) \cap T^{-n}A$ if and only if u can be written as in (3.21), with $T^n(u) \in A = [d, e]$. As T^n restricted to $I(\mathbf{a})$ is continuous and monotone (increasing if n is even, and decreasing if n is odd), it follows that $I(\mathbf{a}) \cap T^{-n}A$ is an interval with endpoints given by

$$\frac{p_n + p_{n-1}d}{q_n + q_{n-1}d}$$

and

$$\frac{p_n + p_{n-1}e}{q_n + q_{n-1}e}.$$

Thus the Lebesgue measure of $I(\mathbf{a}) \cap T^{-n}A$,

$$\left| \frac{p_n + p_{n-1}d}{q_n + q_{n-1}d} - \frac{p_n + p_{n-1}e}{q_n + q_{n-1}e} \right|,$$

expands to

$$\left| \frac{(p_n + p_{n-1}d)(q_n + q_{n-1}e) - (p_n + p_{n-1}e)(q_n + q_{n-1}d)}{(q_n + q_{n-1}d)(q_n + q_{n-1}e)} \right|$$

$$= \left| \frac{p_n q_{n-1}e + p_{n-1}q_n d - p_n q_{n-1}d - p_{n-1}q_n e}{(q_n + q_{n-1}d)(q_n + q_{n-1}e)} \right|$$

$$= (e-d)\frac{|p_n q_{n-1} - p_{n-1}q_n|}{(q_n + q_{n-1}e)(q_n + q_{n-1}f)} = (e-d)\frac{1}{(q_n + q_{n-1}e)(q_n + q_{n-1}f)}$$

by (3.8). On the other hand, the Lebesgue measure of $I(\mathbf{a})$ is

$$\left| \frac{p_n}{q_n} - \frac{p_n + p_{n-1}}{q_n + q_{n-1}} \right| = \frac{|p_n q_{n-1} - p_{n-1}q_n|}{q_n(q_n + q_{n-1})} = \frac{1}{q_n(q_n + q_{n-1})} \tag{3.23}$$

again by (3.8), which implies that

$$\begin{aligned} m(I(\mathbf{a}) \cap T^{-n}A) &= m(A)m(I(\mathbf{a}))\frac{q_n(q_n + q_{n-1})}{(q_n + q_{n-1}e)(q_n + q_{n-1}f)} \\ &\asymp m(A)m(I(\mathbf{a})), \end{aligned} \tag{3.24}$$

where m denotes Lebesgue measure on $(0,1)$. Next notice that

$$\frac{m(B)}{2\log 2} \leqslant \mu(B) \leqslant \frac{m(B)}{\log 2}$$

for any Borel set $B \subseteq (0,1)$, which together with (3.24) gives (3.22).

Now assume that $A \subseteq (0,1)$ is a Borel set with $T^{-1}A = A$. For such a set, the estimate in (3.22) reads as

$$\mu(A \cap I(\mathbf{a})) \asymp \mu(A)\mu(I(\mathbf{a}))$$

for any interval $I(\mathbf{a})$ defined by $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{N}^n$ and any n. However, for a fixed n the intervals $I(\mathbf{a})$ partition $(0,1)$ (as $\mathbf{a}$ varies in $\mathbb{N}^n$), and by (3.23)

$$\begin{aligned} \operatorname{diam}(I(\mathbf{a})) &= \frac{1}{q_n(q_n + q_{n-1})} \\ &\leqslant \frac{1}{2^{n-2}} \qquad \text{(by (3.6))}, \end{aligned}$$

so the lengths of the sets in this partition shrink to zero uniformly as $n \to \infty$. Therefore, the intervals $I(\mathbf{a})$ generate the Borel σ-algebra, and so

$$\mu(A \cap B) \asymp \mu(A)\mu(B)$$

for any Borel subset $B \subseteq (0,1)$ (again by Theorem A.4). We apply this to the set $B = (0,1) \smallsetminus A$ and obtain $0 \asymp \mu(A)\mu(B)$, which shows that either $\mu(A) = 0$ or $\mu((0,1) \smallsetminus A) = 0$, as needed. □

We will use the ergodicity of the Gauss map in Corollary 3.8 to deduce statements about the digits of the continued fraction expansion of a typical real number. Just as Borel's normal number theorem (Example 1.2) gives precise statistical information about the decimal expansion of almost every real number, ergodicity of the Gauss map gives precise statistical information about the continued fraction digits of almost every real number. Of course the form of the conclusion is necessarily different. For example, since there are infinitely many different digits in the continued fraction expansion, they cannot all occur with equal frequency, and (3.25) makes precise the way in which small digits occur more frequently than large ones. We also obtain information on the geometric and arithmetic mean of the digits a_n in (3.26) and (3.27), the growth rate of the denominators q_n in (3.28), and the rate at which the convergents $\frac{p_n}{q_n}$ approximate a typical real number in (3.29).

In particular, equations (3.28) and (3.29) together say that the digit a_{n+1} appearing in the estimate (3.14) does not affect the logarithmic rate of approximation of an irrational by the continued fraction partial quotients significantly.

Corollary 3.8. *For almost every real number $x = [a_1, a_2, \dots] \in (0,1)$, the digit j appears in the continued fraction with density*

$$\frac{2\log(1+j) - \log j - \log(2+j)}{\log 2}, \tag{3.25}$$

$$\lim_{n\to\infty}(a_1a_2\dots a_n)^{1/n}=\prod_{a=1}^{\infty}\left(\frac{(a+1)^2}{a(a+2)}\right)^{\log a/\log 2}, \tag{3.26}$$

$$\lim_{n\to\infty}\frac{1}{n}(a_1+a_2+\cdots+a_n)=\infty, \tag{3.27}$$

$$\lim_{n\to\infty}\frac{1}{n}\log q_n(x)=\frac{\pi^2}{12\log 2}, \tag{3.28}$$

and

$$\lim_{n\to\infty}\frac{1}{n}\log\left|x-\frac{p_n(x)}{q_n(x)}\right|\longrightarrow-\frac{\pi^2}{6\log 2}. \tag{3.29}$$

Proof. The digit j appears in the first N digits with frequency

$$\begin{aligned}\frac{1}{N}|\{i\mid i\leqslant N, a_i=j\}| &= \frac{1}{N}|\{i\mid i\leqslant N, T^ix\in(\tfrac{1}{j+1},\tfrac{1}{j})\}|\\ &\to\frac{1}{\log 2}\int_{1/(j+1)}^{1/j}\frac{1}{1+y}\,\mathrm{d}y\\ &= \frac{2\log(1+j)-\log j-\log(2+j)}{\log 2},\end{aligned}$$

which proves (3.25).

Define a function f on $(0,1)$ by $f(x)=\log a$ for $x\in\left(\frac{1}{a+1},\frac{1}{a}\right)$. Then

$$\begin{aligned}\int_0^1 f(x)\,\mathrm{d}x &= \sum_{a=1}^{\infty}\left(\frac{1}{a}-\frac{1}{a+1}\right)\log a\\ &\leqslant\sum_{a=1}^{\infty}\frac{1}{a^2}\log a<\infty,\end{aligned}$$

so $\int_0^1 f\,\mathrm{d}\mu<\infty$ also, since the density $\frac{\mathrm{d}\mu}{\mathrm{d}x}=\frac{1}{(1+x)\log 2}$ is bounded on $[0,1]$. By the pointwise ergodic theorem (Theorem 2.30) we therefore have, for almost every x,

$$\frac{1}{n}\sum_{j=0}^{n-1}\log a_j=\frac{1}{n}\sum_{j=0}^{n-1}f(T^jx)\longrightarrow\int f(x)\,\mathrm{d}\mu.$$

This shows (3.26) since

$$\int_0^1 f\,\mathrm{d}\mu=\sum_{a=1}^{\infty}\frac{\log a}{\log 2}\int_{1/(1+a)}^{1/a}\frac{1}{1+x}\,\mathrm{d}x.$$

Now consider the function $g(x) = \mathrm{e}^{f(x)}$ (so $g(x) = a_1$ is the first digit in the continued fraction expansion of x). We have

$$\frac{1}{n}(a_1 + \cdots + a_n) = \frac{1}{n}\sum_{j=0}^{n-1} g(T^j x),$$

but the pointwise ergodic theorem cannot be applied to g since $\int_0^1 g\,\mathrm{d}\mu = \infty$ (the result needed is Exercise 2.6.5(2); the argument here shows how to do this exercise). However, for any fixed N the truncated function

$$g_N(x) = \begin{cases} g(x) & \text{if } g(x) \leqslant N; \\ 0 & \text{if not} \end{cases}$$

is in L^1_μ since

$$\int g_N\,\mathrm{d}\mu = \frac{1}{\log 2}\sum_{a=1}^{N}\int_{1/(a+1)}^{1/a} a\,\mathrm{d}x = \frac{1}{\log 2}\sum_{a=1}^{N}\frac{1}{a+1}.$$

Notice that $\int_0^1 g_N\,\mathrm{d}\mu \to \infty$ as $N \to \infty$. By the ergodic theorem,

$$\begin{aligned}\liminf_{n\to\infty}\frac{1}{n}\sum_{j=0}^{n-1} g(T^j x) &\geqslant \lim_{n\to\infty}\frac{1}{n}\sum_{j=0}^{n-1} g_N(T^j x)\\ &= \int_0^1 g_N\,\mathrm{d}\mu \to \infty\end{aligned}$$

as $N \to \infty$, showing (3.27).

The proofs of (3.25) and (3.26) were straightforward applications of the ergodic theorem, and (3.27) only required a simple extension to measurable functions. Proving (3.28) and (3.29) takes a little more effort.

First notice that

$$\begin{aligned}\frac{p_n(x)}{q_n(x)} &= \frac{1}{a_1 + [a_2, \dots, a_n]}\\ &= \frac{1}{a_1 + \frac{p_{n-1}(Tx)}{q_{n-1}(Tx)}}\\ &= \frac{q_{n-1}(Tx)}{p_{n-1}(Tx) + q_{n-1}(Tx)a_1},\end{aligned}$$

so $p_n(x) = q_{n-1}(Tx)$ since the convergents are in lowest terms. Recall that we always have $p_1 = q_0 = 1$. It follows that

$$\frac{1}{q_n(x)} = \frac{p_n(x)}{q_n(x)} \cdot \frac{p_{n-1}(Tx)}{q_{n-1}(Tx)} \cdots \frac{p_1(T^{n-1}x)}{q_1(T^{n-1}x)},$$

so

$$-\frac{1}{n}\log q_n(x) = \frac{1}{n}\sum_{j=0}^{n-1}\log\left[\frac{p_{n-j}(T^j x)}{q_{n-j}(T^j x)}\right].$$

Let $h(x) = \log x$ (so $h \in L^1_\mu$). Then

$$-\frac{1}{n}\log q_n(x) = \frac{1}{n}\underbrace{\sum_{j=0}^{n-1} h(T^j x)}_{S_n} - \frac{1}{n}\underbrace{\sum_{j=0}^{n-1}\left[\log(T^j x) - \log\left(\frac{p_{n-j}(T^j x)}{q_{n-j}(T^j x)}\right)\right]}_{R_n}$$

gives a splitting of $-\frac{1}{n}\log q_n(x)$ into an ergodic average $S_n = \mathsf{A}^n_h$ and a remainder term R_n. By the ergodic theorem,

$$\lim_{n\to\infty}\frac{1}{n}S_n = \frac{1}{\log 2}\int_0^1 \frac{\log x}{1+x}\,\mathrm{d}x = -\frac{\pi^2}{12\log 2}.$$

To complete the proof of (3.28), we need to show that $\frac{1}{n}R_n \to 0$ as $n \to \infty$. This will follow from the observation that $\frac{p_{n-j}(T^j x)}{q_{n-j}(T^j x)}$ is a good approximation to $T^j x$ if $(n-j)$ is large enough. Recall from (3.7) and (3.6) that

$$p_k \geqslant 2^{(k-2)/2},\ q_k \geqslant 2^{(k-1)/2},$$

so, by using the inequality (3.13),

$$\left|\frac{x}{p_k/q_k} - 1\right| = \frac{q_k}{p_k}\left|x - \frac{p_k}{q_k}\right| \leqslant \frac{1}{p_k q_{k+1}} \leqslant \frac{1}{2^{k-1}}.$$

By using this together with the fact that $|\log u| \leqslant 2|u-1|$ whenever $u \in [\frac{1}{2}, \frac{3}{2}]$ (which applies in the sum below with $j \leqslant n-2$), we get

$$\begin{aligned}|R_n| &\leqslant \sum_{j=0}^{n-1}\left|\log\frac{T^j x}{p_{n-j}(T^j x)/q_{n-j}(T^j x)}\right| \\ &\leqslant \underbrace{2\sum_{j=0}^{n-2}\left|\frac{T^j x}{p_{n-j}(T^j x)/q_{n-j}(T^j x)} - 1\right|}_{T_n} + \underbrace{\left|\log\frac{T^{n-1}x}{p_1(T^{n-1}x)/q_1(T^{n-1}x)}\right|}_{U_n}.\end{aligned}$$

Now

$$T_n \leqslant \sum_{j=0}^{n-2}\frac{2}{2^{n-j-1}} \leqslant 2$$

for all n. For the second term, notice that

$$U_n = \left|\log\left[\left(T^{n-1}x\right)a_1\left(T^{n-1}x\right)\right]\right|,$$

and by the inequality (3.17) we have

$$1 \geqslant \left(T^{n-1}x\right)a_1\left(T^{n-1}x\right) \geqslant \frac{a_1\left(T^{n-1}x\right)}{1+a_1\left(T^{n-1}x\right)} \geqslant \frac{1}{2}$$

since $a_1(T^{n-1}x) \geqslant 1$. Therefore,

$$|\log\left[\left(T^{n-1}x\right)a_1\left(T^{n-1}x\right)\right]| \leqslant \log 2,$$

which completes the proof that

$$\frac{1}{n}R_n \to 0$$

as $n \to \infty$, and hence shows (3.28).

Equation (3.29) follows from (3.28), since from the inequalities (3.13) and (3.15) we have

$$\log q_n + \log q_{n+1} \leqslant -\log\left|x - \frac{p_n}{q_n}\right| \leqslant \log q_n + \log q_{n+2}.$$

□

Exercises for Sect. 3.2

Exercise 3.2.1. Use the idea in the proof of (3.27) to extend the pointwise ergodic theorem (Theorem 2.30) to the case of a measurable function $f \geqslant 0$ with $\int_X f\,\mathrm{d}\mu = \infty$ without the assumption of ergodicity.

Exercise 3.2.2. Show that the map from $\mathbb{N}^{\mathbb{N}}$ to $[0,1]\smallsetminus\mathbb{Q}$ sending $(a_1, a_2, \dots)$ to $[a_1, a_2, \dots]$ is a homeomorphism with respect to the discrete topology on $\mathbb{N}$ and the product topology on $\mathbb{N}^{\mathbb{N}}$.

Exercise 3.2.3. Let $\mathbf{p} = (p_1, p_2, \dots)$ be an infinite probability vector (this means that $p_i \geqslant 0$ for all i, and $\sum_i p_i = 1$). Show that $\mathbf{p}$ gives rise to a σ-invariant and ergodic probability measure $\mathbf{p}^{\mathbb{N}}$ on $\mathbb{N}^{\mathbb{N}}$.

Exercise 3.2.4. Let $\phi : \mathbb{N}^{\mathbb{N}} \to (0,1)\smallsetminus\mathbb{Q}$ be the map discussed on page 79, and let μ be the Gauss measure on $[0,1]$. Show that $\phi_*^{-1}\mu$ is not of the form $\mathbf{p}^{\mathbb{N}}$ for any infinite probability vector $\mathbf{p}$.